

AMITH LAL

Generative AI Engineer · RAG & LLM Systems

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PROFESSIONAL SUMMARY

Generative AI Engineer shipping production **RAG** systems on self-hosted LLMs: hybrid retrieval, **LoRA/QLoRA** fine-tuning, prompt testing, and **RAGAS** evaluation, delivered through **Django/FastAPI** APIs integrated with **SAP**. Strong **Python** and clean, secure, maintainable code; **AWS** certified, with solid SQL, ML, Docker, Git, and CI/CD.

CORE COMPETENCIES

Gen AI Stack	Local LLMs (Qwen, Llama) · vLLM · Vector DBs (Qdrant, ChromaDB, FAISS) · BGE-M3 · LangChain · LangGraph · n8n · RAG · Hybrid Search · Corrective Self-RAG · RAGAS · NeMo Guardrails · Presidio · Prompt Design & Testing · Agentic Workflows · Claude Code · MCP Servers
Machine Learning	Scikit-learn · LoRA/QLoRA Fine-tuning · Predictive Modeling · Model Deployment · Data Pipelines
Languages	Python (strong) · SQL · Java
Backend & APIs	Django · Django REST Framework · FastAPI · RESTful APIs · API Design & Maintenance · Streamlit · System Architecture · Firebase
DevOps & Cloud	AWS (EC2, S3, IAM) – AWS Certified · Vercel · Docker · Kubernetes · CI/CD · Git / GitHub · Code Reviews
Databases & Storage	Supabase (PostgreSQL) · MySQL · SQLite · Firestore
Engineering Practices	Clean, Secure & Maintainable Code · OOP · Data Structures & Algorithms · API Integration · Postman

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineer

Jul 2024 – Present

Linx AS, Kochi, India

Gen AI Document Processing & Extraction Platform | Python, Django, vLLM, LLMs, RAG

- Developed Django-based **Generative AI** applications using **LangChain**, **LangGraph**, and **RAG** to orchestrate **self-hosted LLMs (Qwen, Llama)** served with **vLLM**, ingesting and extracting structured data from diverse documents at **~400 documents/week** with **~95% field-level accuracy**, cutting handling time from **10–20 min to ~2 min**.
- Implemented **RAG pipelines** end to end: document chunking, embedding generation, and **hybrid** vector search (**Qdrant**, **ChromaDB**) powering Q&A, AI summaries, and targeted extraction; added a **Corrective Self-RAG loop** to catch weak retrievals before generation.
- Evaluated retrieval and generation quality with **RAGAS** (context precision, faithfulness, answer relevancy) plus hit rate and MRR; designed **schema-constrained JSON** outputs for direct downstream integration, with **guardrails (NeMo Guardrails)** and **confidence-based human review** before submission.
- Fine-tuned **Llama** models with **QLoRA** for query rewriting and chunk grading, improving domain accuracy and reducing inference cost; orchestrated **AI agents with tool-calling workflows**.
- Configured and integrated **MCP servers** to give LLM agents structured tool access; designed and iteratively **tested LLM prompts** to improve extraction accuracy and response reliability.
- Built and maintained **RESTful APIs** with **Django REST Framework** and **FastAPI** for file uploads and extraction, owning system architecture and integrating directly with **SAP**; used **Claude Code** for AI-assisted development, with clean, secure code, Git best practices, and **code reviews**.

Albert Invent Data Migration & Utility Development | Python, Albert SDK, Pandas, Streamlit

- Migrated critical **R&D data** from client legacy systems into the **Albert Invent** platform, prioritizing data integrity and minimal downtime.
- Automated extraction, validation, and loading with **Python and the Albert SDK**, using bulk-processing scripts to make migrations repeatable across clients.
- Built a **Streamlit** app for validation checks and real-time migration status tracking.

Cloud Engineering Intern

Feb 2024 – May 2024

Trainso Training Solutions, Bengaluru, India

- Worked on **AWS** operations (IAM, EC2, S3, security configurations); assisted in deploying **Docker** applications and managing access control policies.

EDUCATION

B.Tech – Computer Science · APJ Abdul Kalam Technological University · CGPA: 9.34

2019 – 2023

CERTIFICATIONS & PUBLICATIONS

ML Specialization, Andrew Ng (2024) · IBM Applied AI Professional Certificate (2023) · Python for Data Science & AI, IBM (2023) · AWS Solutions Architect Associate (2023) · Publication: Personalizable Super-App Framework, NCASCD (2023)